

Distance Pipeline — Cycle-by-Cycle at End of Feature Vector

Three registered stages in the distance accumulator:

Stage 1: $\text{diff_reg} \leftarrow x[i] - \text{sv}[i]$ (subtract)
Stage 2: $\text{sq_reg} \leftarrow \text{diff_reg}^2$ (square)
Stage 3: $\text{acc} \leftarrow \text{acc} + \text{sq_reg}$ (accumulate)

Each stage is clocked, so after the last feature index ($i = 255$) two drain cycles are needed to flush diff_reg and sq_reg through to acc .

Cycle Table

Cycle	dim_idx	diff_reg	sq_reg	acc update
1	0	$\delta[0]$	—	—
2	1	$\delta[1]$	$\delta[0]^2$	—
3	2	$\delta[2]$	$\delta[1]^2$	$D += \delta[0]^2$
...	...	...	...	...
252	251	$\delta[251]$	$\delta[250]^2$	$D += \delta[249]^2$
253	252	$\delta[252]$	$\delta[251]^2$	$D += \delta[250]^2$
254	253	$\delta[253]$	$\delta[252]^2$	$D += \delta[251]^2$
255	254	$\delta[254]$	$\delta[253]^2$	$D += \delta[252]^2$
256	255	$\delta[255]$	$\delta[254]^2$	$D += \delta[253]^2$
257	drain 1	—	$\delta[255]^2$	$D += \delta[254]^2$
258	drain 2	—	—	$D += \delta[255]^2$ ✓

D is valid after cycle 258. FSM holds ACCUMULATE for cycles 257–258 before transitioning to KERNEL.

Why 2 Drain Cycles

After dim_idx hits 255, two values are still in-flight in registered stages:

- `diff_reg` holds $\delta[255]$ — needs one cycle to become `sq_reg`
- `sq_reg` will hold $\delta[255]^2$ — needs one more cycle to reach `acc`

Skipping the drain (v6 bug) caused accuracy collapse because the last two squared differences were never accumulated into D.

General Formula

LAT	Cycles per SV
1	$256 \times 1 + 2 = \mathbf{258}$
2	$256 \times 2 + 2 = \mathbf{514}$
3	$256 \times 3 + 2 = \mathbf{770}$

LAT = SRAM read latency (number of cycles to fetch one feature word). The +2 is always the fixed pipeline drain, independent of LAT.